
THE THIRD INVOLUTION

$x \rightarrow x/(x-1)$ · a projective construction

Working note · May 2026

I. THE SIX TRANSFORMATIONS

The projective line possesses exactly six Möbius transformations that permute the three special points $\{0, 1, \infty\}$ among themselves. A Möbius transformation is completely determined by the images of three points, and three points admit exactly $3! = 6$ permutations. The six transformations correspond to the six elements of the symmetric group S_3 , known in this context as the *anharmonic group*.

Transformation	Formula	Action on $\{0, 1, \infty\}$	Order
Identity	x	fixes all three	1
Complement	$1 - x$	fixes ∞ , swaps $0 \leftrightarrow 1$	2
Reciprocal	$1/x$	fixes 1, swaps $0 \leftrightarrow \infty$	2
Third involution	$x/(x-1)$	fixes 0, swaps $1 \leftrightarrow \infty$	2
Nested complement	$1/(1-x)$	$0 \rightarrow 1 \rightarrow \infty \rightarrow 0$	3
Nested reciprocal	$(x-1)/x$	$0 \rightarrow \infty \rightarrow 1 \rightarrow 0$	3

The three order-2 elements (involutions) are shown; the third involution is highlighted.

II. THE ASYMMETRY AMONG THE THREE INVOLUTIONS

The group-theoretic account places all three involutions on equal footing — each fixes one of $\{0, 1, \infty\}$ and swaps the other two. But geometrically the three points are not equivalent, and the involutions that fix them reflect this.

Complement (fixes ∞ , swaps $0 \leftrightarrow 1$) operates within the additive world. The unit interval $[0, 1]$ is its natural habitat. The name is self-explanatory.

Reciprocal (fixes 1, swaps $0 \leftrightarrow \infty$) operates within the multiplicative world. The unit 1 is the fixed point because $1/1 = 1$.

The third involution (fixes 0, swaps $1 \leftrightarrow \infty$) is different in kind. It fixes the origin — the point prior to both addition and multiplication — and exchanges the unit with the horizon. It does not operate *within* either register; it operates *between* them.

III. THE PROJECTIVE CONSTRUCTION

Complement and Reciprocal are defined analytically — each is an operation on a value. The third involution admits a purely geometric definition, prior to any formula.

Place two copies of the positive real line as coordinate axes meeting at the origin 0. The two unit points determine a single point in the plane: the hinge (1, 1), corner of the unit square.

Construction. Given x on the horizontal axis ($x > 1$), draw the straight line through x and the hinge (1, 1). This line meets the vertical axis at $y = x/(x-1)$.

Verification: the line through $(x, 0)$ and $(1, 1)$ parameterizes as $(x + t(1-x), t)$. Setting the first coordinate to zero gives $t = x/(x-1)$, which is the vertical intercept. The formula is what the straight line computes.

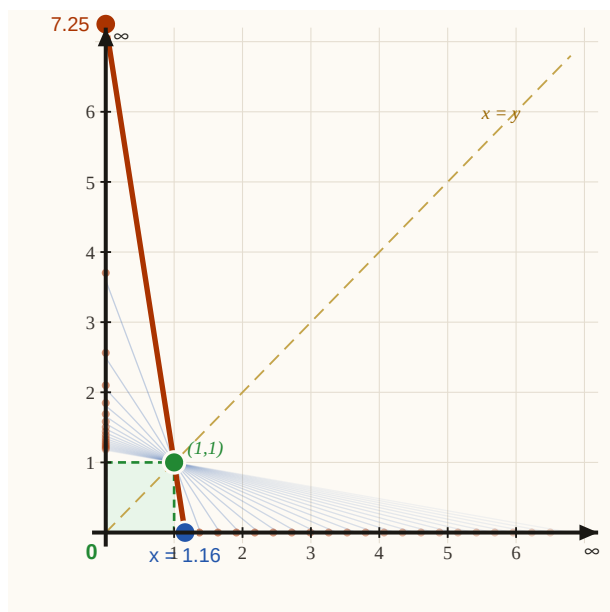


Fig. 1 — $x \approx 1.16$, $f(x) \approx 7.1$ (approaching 1, value escapes to ∞)

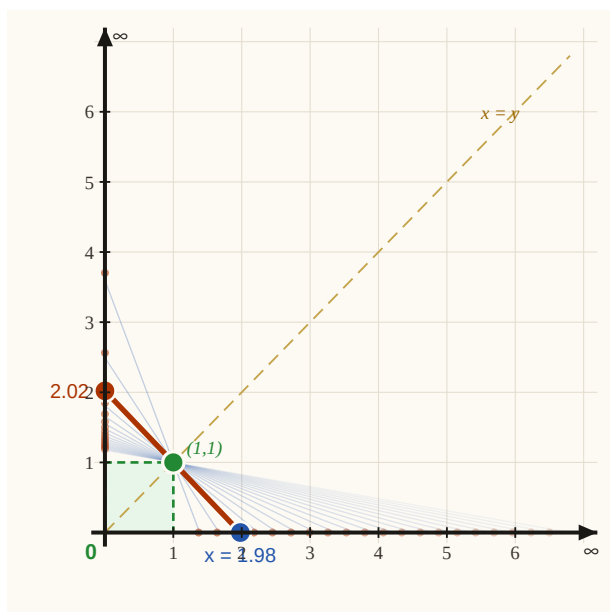


Fig. 2 — $x \approx 1.98$, $f(x) \approx 2.02$ (near fixed point $x = 2$)

IV. WHAT THE CONSTRUCTION SHOWS

The involution is visible. The line through $(x, 0)$ and $(1, 1)$ is the same line as the line through $(0, y)$ and $(1, 1)$. Applying the construction twice returns to the start. The symmetry is built in, not appended.

The fixed point $x = 2$ is visible (Fig. 2). The fixed points satisfy $x(x-2) = 0$, giving $x = 0$ or $x = 2$. Geometrically, $x = 2$ is where the active line meets the diagonal $y = x$ — the line is at 45° and horizontal and vertical values coincide.

The exchange $1 \leftrightarrow \infty$ is visible (Fig. 1). As x approaches 1 from the right, the line through $(x, 0)$ and the hinge $(1, 1)$ flattens toward vertical and the intercept escapes to ∞ . As x grows without bound, the intercept descends toward 1. The pencil enacts the exchange.

The origin is anchor, not participant. Every line passes through the hinge $(1, 1)$. The origin 0 is fixed not because anything happens there, but because it is where both axes begin — the common ground the construction rests on.

V. MINIMAL APPARATUS: THREE CONSTRUCTIONS COMPARED

The three involutions can each be constructed geometrically, and the tools each construction requires reveal a genuine structural difference among them. The comparison is not about which construction is more complex but about what each one minimally needs in order to exist.

Complement $x \rightarrow 1-x$ requires one line with two marked points, 0 and 1. The construction is midpoint bisection: find the point $\frac{1}{2}$ equidistant from both ends, then reflect any point x across that midpoint. This is a compass construction — it measures distance and bisects a segment. No multiplication, no ratio, no second axis. The unit is treated as a *length*: the distance from 0 to 1.

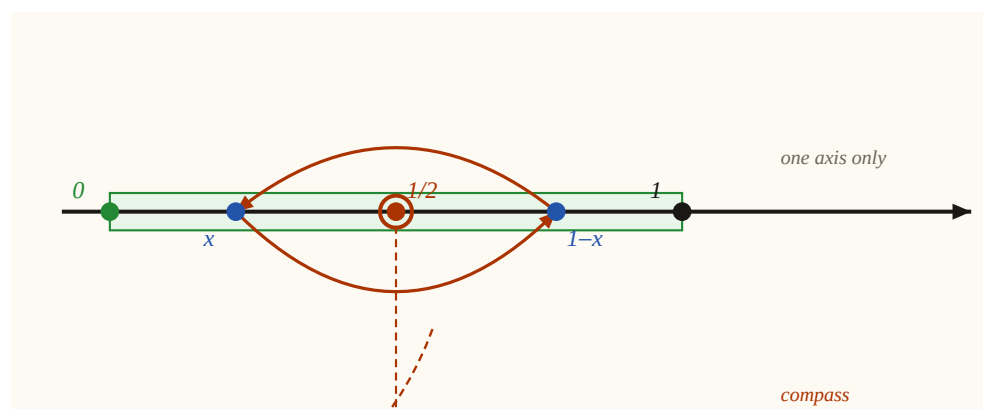


Fig. 5 — Complement: one line, compass bisection. No second axis, no multiplication.

Reciprocal $x \rightarrow 1/x$ requires one line with one marked point: the unit 1. The construction uses harmonic conjugates — a classical projective technique requiring only a straightedge. Given points -1 , 1, and x on the line, two auxiliary points S and T and a network of straightedge lines determine $1/x$ as the harmonic conjugate of x with respect to $\{-1, 1\}$. This works because -1 and 1 are precisely the fixed points of the Reciprocal transformation: the construction uses the

transformation's own fixed points as its projective reference. No compass, no distance measurement, no second axis. The unit is treated as a *ratio*: the multiplicative identity.

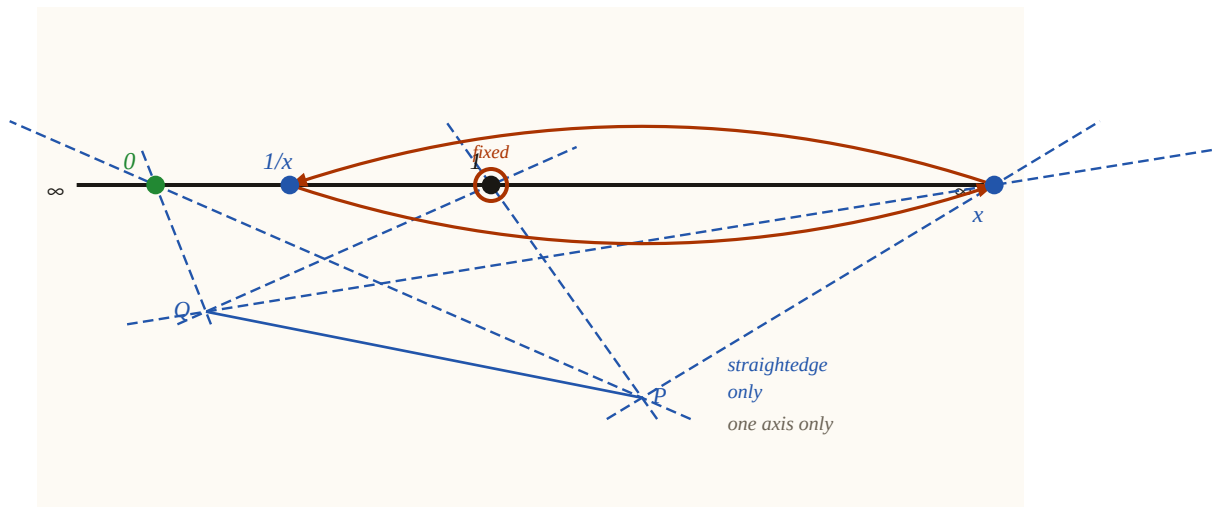


Fig. 6 — Reciprocal: one line, harmonic conjugate construction. No compass, no second axis.

The contrast is classical: the compass belongs to the metric, additive world of distance and length; the straightedge belongs to the projective, multiplicative world of ratios and cross-ratios. Complement needs the compass. Reciprocal needs only the straightedge. Each is complete within its own world.

The third involution $x \rightarrow x/(x-1)$ requires both. Its hinge at $(1, 1)$ cannot be located from either axis alone. The full demonstration follows in Sections VI and VII.

VI. THE HYPERBOLA AND THE ORIGIN OF THE HINGE

The projective construction pivots on the hinge at $(1, 1)$. The question left open above is: where does that point come from? It was introduced as the corner of the unit square, which is true but insufficient. A stronger account is available, and it is geometric.

Consider the unit hyperbola $xy = 1$. This curve is the locus of all reciprocal pairs — every point on it has the form $(x, 1/x)$. It is asymptotic to both axes, touching neither, defined by both simultaneously. It encodes the Reciprocal transformation as a geometric object: the reflection across the diagonal $x = y$ maps the hyperbola to itself, because swapping coordinates sends $(x, 1/x)$ to $(1/x, x)$, which is also on the hyperbola.

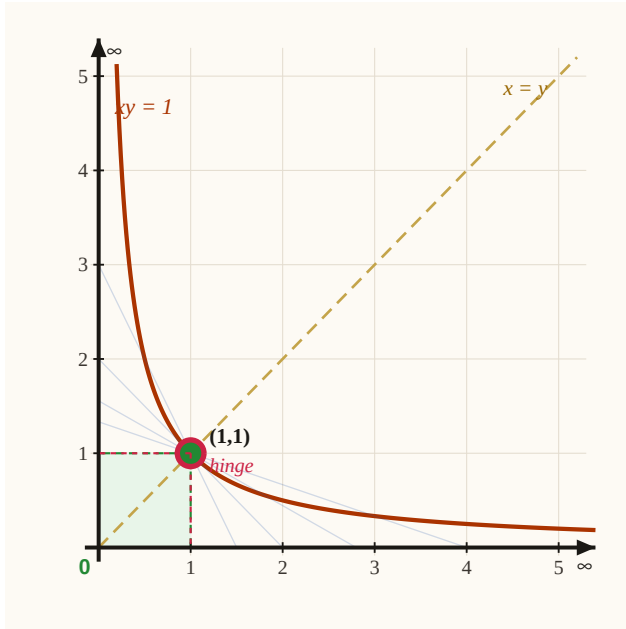


Fig. 4 — The unit hyperbola $xy = 1$ meets the diagonal at $(1,1)$: the hinge.

The diagonal $x = y$ is the locus of points where the two coordinates are equal — where input and output, horizontal and vertical, are in balance. Its intersection with the hyperbola $xy = 1$ is the unique point where a value and its reciprocal coincide:

$$x = 1/x \rightarrow x^2 = 1 \rightarrow x = 1$$

That intersection is $(1, 1)$. It is not chosen — it is *found*. The hyperbola and the diagonal together force it into existence.

This point is the hinge of the third involution. The pencil of lines through $(1, 1)$ that defines $x/(x-1)$ is therefore not an arbitrary construction — it is the pencil through the point the unit hyperbola selects when asked: where do the two registers meet at equal value?

VII. WHAT THE HYPERBOLA DEMONSTRATES

The hyperbola argument establishes a chain of geometric necessity:

Two axes with a shared unit	the basic apparatus of measurement in two registers
Reciprocal correspondence $xy = 1$	the curve encoding the relationship between the two registers
Intersection with diagonal $x = y$	the point where the two registers are in balance: $(1, 1)$
Pencil of lines through $(1, 1)$	the projective construction of $x/(x-1)$
Third involution $x/(x-1)$	fixes 0 and 2, swaps $1 \leftrightarrow \infty$

Each step is forced by the previous one. No step imports vocabulary from the Stick/Turn framework — the chain runs entirely within geometry.

Now compare the three involutions against this chain:

Complement $x \rightarrow 1-x$ requires one axis and a unit marked on it. It reflects within the unit interval. It does not need the second axis, the hyperbola, or the diagonal.

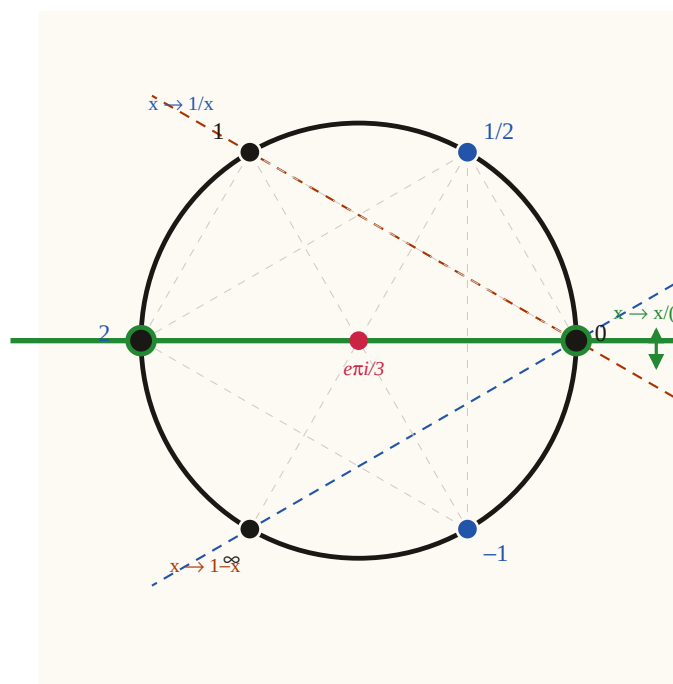
Reciprocal $x \rightarrow 1/x$ requires one axis and a unit marked on it. It inverts within the multiplicative structure of a single line. It does not need the second axis — though the hyperbola encodes its action when both axes are present.

The third involution $x \rightarrow x/(x-1)$ cannot be constructed from either axis alone. Its hinge is produced by the hyperbola — which requires both axes — intersecting the diagonal. Remove either axis and the hyperbola vanishes. Remove the hyperbola and the hinge vanishes. Remove the hinge and the construction has no content.

This is the demonstration. The third involution is structurally dependent on the joint presence of both registers in a way that the other two are not. That dependence is not asserted from the framework — it is read off the construction. The geometry requires both axes before the third involution can exist.

VIII. TWO REPRESENTATIONS OF THE SAME STRUCTURE

The circle diagram of the anharmonic group (Fig. 3) and the projective construction (Figs. 1–2) are two representations of the same underlying involution. Reading them together makes the structure visible from both sides.



In the circle diagram, the three involution axes each connect the two points that a given involution swaps. The blue axis connects 0 and ∞ (swapped by Reciprocal). The red axis connects 0 and 1 (swapped by Complement). The green axis connects **0 and 2** — the fixed points of $x/(x-1)$.

These are not merely the endpoints of a diameter. They are the two points that the third involution leaves unmoved — its fixed-point set. The axis is the geometric locus of that stillness.

In the projection diagram, 0 and 2 appear in exactly the same roles. The origin 0 is the anchor of the entire construction — it never participates in any exchange. The value $x = 2$ is where the active line meets the diagonal $y = x$: the balance point, where horizontal and vertical values coincide.

The circle diagram is the projective view: points on the Riemann sphere, axes as diameters, symmetry as rotation. The projection diagram is the Cartesian view: axes as coordinate lines, the hinge as a point in the plane, the transformation as a pencil of lines. The fixed points $\{0, 2\}$ appear in both, unchanged.

The correspondence is exact and not accidental. The green axis in the circle diagram does not merely gesture at 0 and 2 as interesting values — it is the fixed-point set of $x/(x-1)$, drawn as a geometric object. What the algebra computes, the geometry draws. The two diagrams are one statement.

IX. ALL THREE INVOLUTIONS ACTING SIMULTANEOUSLY

The diagram below places all three involutions on the same coordinate system, acting on the same value of x at once. The structural difference established in Sections V–VII is now directly visible rather than described.

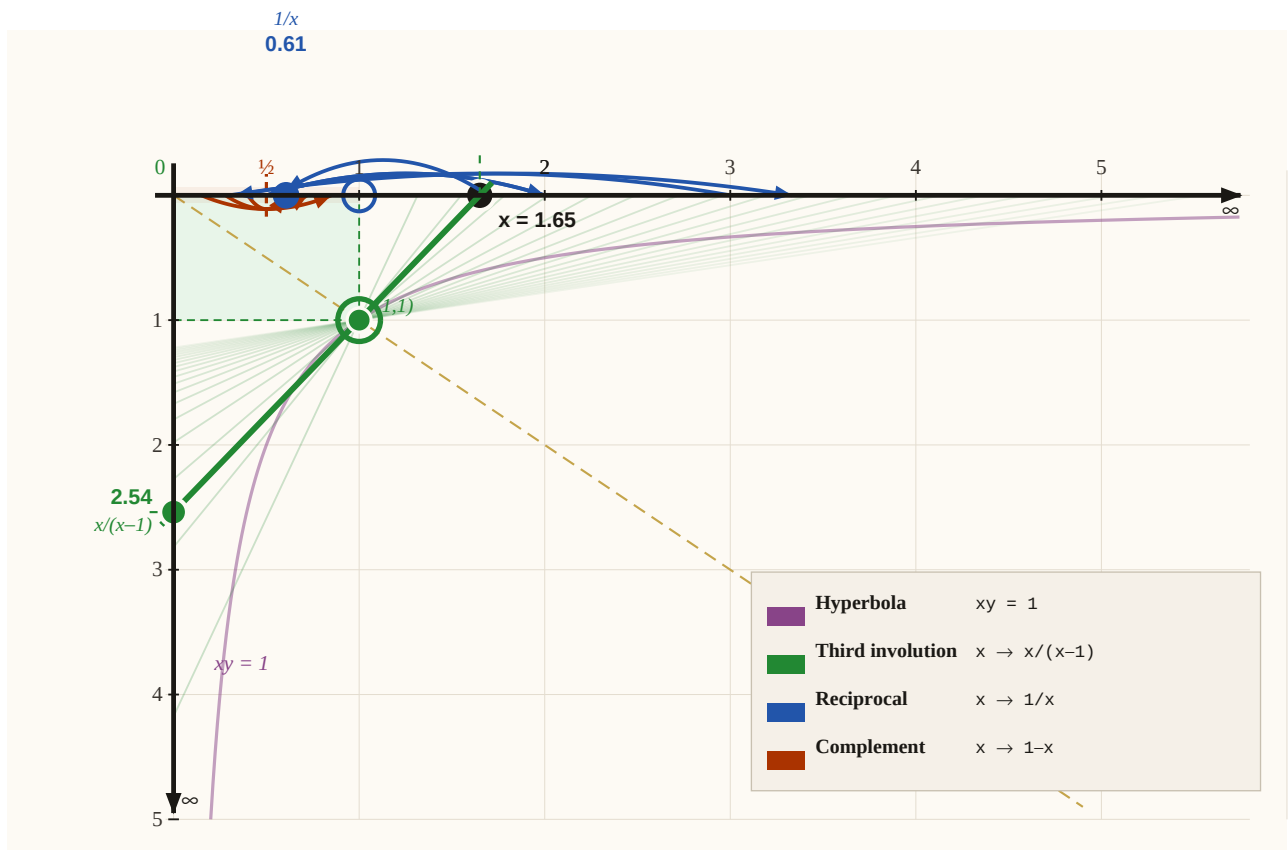


Fig. 7 — All three involutions acting on $x = 1.65$ simultaneously. Red: above axis. Blue: below axis. Green: crosses to vertical axis. Red arcs above the axis show Complement acting within the unit interval. The paired points swap across the fixed midpoint $\frac{1}{2}$. The construction never leaves the horizontal axis and never ventures beyond the unit segment.

Blue arcs below the axis show Reciprocal acting along the full positive line. The paired points swap across the fixed point 1, ranging from near-zero to large values. The construction also stays on the horizontal axis, but its range is unbounded.

The green line shows the third involution: a single straight line through the hinge $(1, 1)$ connecting x on the horizontal axis to $x/(x-1)$ on the vertical axis. It is the only transformation of the three that leaves the line. Its result lives on a different axis entirely.

The purple hyperbola $xy = 1$ passes through the hinge, reminding us where that point comes from. Complement and Reciprocal need no hyperbola. The third involution cannot exist without it.

X. THE COMPLETE FIXED-POINT STRUCTURE

The six transformations of the anharmonic group have a complete and symmetric fixed-point structure. Each transformation is fully characterized by its fixed points, and the pattern across the

group is worth examining as a whole.

Transformation	Formula	Fixed points	Where
Identity	x	every point	real
Complement	$1 - x$	$\{1/2, \infty\}$	real
Reciprocal	$1/x$	$\{1, -1\}$	real
Third involution	$x/(x-1)$	$\{0, 2\}$	real
Reach-leading	$1/(1-x)$	$e^{(\pm i \pi/3)}$	complex only
Turn-leading	$(x-1)/x$	$e^{(\pm i \pi/3)}$	complex only

The three involutions each own a distinct pair of real fixed points. The two 3-cycles share one pair of complex fixed points. The split is exact: real fixed points for order-2 elements, complex fixed points for order-3 elements.

XI. WHY THE 3-CYCLES HAVE NO REAL FIXED POINTS

The fixed points of the 3-cycles are the roots of $x^2-x+1=0$, which has discriminant -3 and no real solutions. This is not merely an algebraic observation. It has a geometric account.

A fixed point of the composition Complement-then-Reciprocal is a point x such that Complement sends x to some y and Reciprocal sends y back to x . That means $\{x, y\}$ is simultaneously a 2-cycle of Complement and a 2-cycle of Reciprocal.

The 2-cycle condition for each involution is straightforward to derive:

Complement	$1-x = y$ and $1-y = x$	$x + y = 1$
Reciprocal	$1/x = y$ and $1/y = x$	$xy = 1$

A shared 2-cycle must satisfy both conditions simultaneously: $x + y = 1$ and $xy = 1$. These are the elementary symmetric polynomials of the quadratic $t^2-t+1=0$. The discriminant is -3 . No real pair $\{x, y\}$ satisfies both conditions.

The two conditions have a natural reading. $x + y = 1$ says the pair has arithmetic mean $1/2$: it is centered on the additive midpoint of the unit interval. $xy = 1$ says the pair has geometric mean 1: it is balanced around the multiplicative identity. For positive real numbers the arithmetic mean is always greater than or equal to the geometric mean ($AM \geq GM$). Here the arithmetic mean $1/2$ is less than the geometric mean 1, which is impossible for positive reals. No such pair exists on the positive

real line.

This is the geometric account. The composition has no real fixed points because Complement and Reciprocal are structured around two different notions of the unit — the additive midpoint $\frac{1}{2}$ and the multiplicative identity 1 — and no real pair can simultaneously satisfy both.

XII. WHERE THE FIXED POINTS GO

The roots of $x^2-x+1=0$ are $x = (1 \pm i\sqrt{3})/2 = e^{(\pm i \pi)/3}$. These are the primitive sixth roots of unity, sitting at $\pm 60^\circ$ in the complex plane on the unit circle.

They satisfy both conditions — but in the complex plane rather than on the real line. Verify directly:

Sum	$e^{(i \pi)/3} + e^{(-i \pi)/3} = 2 \cos(\pi/3) = 2 \times 1/2 = 1$	$x + y = 1 \checkmark$
Product	$e^{(i \pi)/3} \times e^{(-i \pi)/3} = e^0 = 1$	$xy = 1 \checkmark$

Both conditions are satisfied — but only by leaving the real line. The real line is the domain where $AM \geq GM$ enforces an ordering that makes the two conditions incompatible. The complex plane is where that ordering no longer applies and the two conditions can be jointly satisfied.

The two 3-cycles share these fixed points because they form the unique cyclic subgroup of order 3 in $S^{\blacksquare}$. They are inverses of each other — one cycles $0 \rightarrow 1 \rightarrow \infty \rightarrow 0$ and the other reverses it — and a cyclic group of order 3 has exactly one fixed-point set shared by both non-identity elements.

The equation $x^2-x+1=0$ encodes the incompatibility between the additive and multiplicative registers over the reals. Its roots $e^{\pm i\pi/3}$ are the unique complex points where both conditions are simultaneously satisfied. The quotient ring $\blacksquare[x]/(x^2-x+1)$ is not imported — it is the forced habitat of the fixed points that the two generators jointly require.

The third involution is what you get when you ask: what transformation fixes the ground and exchanges the two registers at their meeting point? The answer is $x/(x-1)$, and it can be drawn with a straightedge.